

Parametric and Semiparametric Copula Model Selection

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Sklar's Theorem

- A function $C : [0, 1]^d \rightarrow [0, 1]$, is called a **copula** if it is a cumulative distribution function (cdf) with uniform marginals on $[0, 1]$.
- **Sklar's Theorem:** Assume that F is the cdf of an **absolutely continuous distribution**. Then there exists a **unique** copula C such that

$$F(x_1, \dots, x_d) = C(F_1(x_1), \dots, F_d(x_d)),$$

where $F_1, \dots, F_d$ are the marginal cdfs of F .

- Sklar's theorem allows us to separate **the dependence structure** from **the structure of the marginal distributions**.
- Since we consider only absolutely continuous distributions, densities can be derived from Sklar's theorem as

$$f(x_1, \dots, x_d) = c(F_1(x_1), \dots, F_d(x_d)) \prod_{l=1}^d f_l(x_l).$$

- We will be interested in selecting the most appropriate copula c .



Fully Parametric Approach

- Assume that the copula C can be parametrized by a real vector θ , and each marginal cdf F_l can be parametrized by a real vector γ_l for $l = 1, \dots, d$. Then the joint density can be written as

$$f(x_1, \dots, x_d; \theta, \gamma) = c(F_1(x_1; \gamma_1), \dots, F_d(x_d; \gamma_d); \theta) \prod_{l=1}^d f_l(x_l; \gamma_l),$$

where $\gamma = (\gamma_1, \dots, \gamma_d)$.

- In statistics, we observe $\mathcal{X}_n = \{\mathbf{X}_i\}_{i=1}^n \stackrel{\text{i.i.d.}}{\sim} f^\circ$, where f° denotes the unknown data-generating density.
- Let $B \in \mathbb{N}$ denote the number of parametric families under consideration. For each $b = 1, \dots, B$, define

$$\mathcal{F}_b = \left\{ f(\cdot; \theta_b, \gamma_{b,1}, \dots, \gamma_{b,d}) : (\theta_b, \gamma_{b,1}, \dots, \gamma_{b,d}) \in \Theta_b \times \Gamma_{b,1} \times \dots \times \Gamma_{b,d} \right\}.$$

- Our goal is to **rank the fitted models** to identify the most suitable one.



MLE-Based Information Criteria

- By fitting each family $\mathcal{F}_b$ to the data $\mathcal{X}_n$, one can obtain the MLEs

$$\left(\hat{\theta}_b, \hat{\gamma}_{b,1}, \dots, \hat{\gamma}_{b,d}\right).$$

- If $f^\circ \in \mathcal{F}_b$, we compute the Akaike Information Criterion:

$$\text{AIC}_{\text{MLE}}(b) = 2 \cdot \left[\ell \left(\hat{\theta}_b, \hat{\gamma}_{b,1}, \dots, \hat{\gamma}_{b,d} \right) - \dim(\theta_b, \gamma_{b,1}, \dots, \gamma_{b,d}) \right].$$

- If $f^\circ \notin \mathcal{F}_b$, we use the Takeuchi Information Criterion:

$$\text{TIC}_{\text{MLE}}(b) = 2 \cdot \left[\ell \left(\hat{\theta}_b, \hat{\gamma}_{b,1}, \dots, \hat{\gamma}_{b,d} \right) - \widehat{\text{bias}}_{\text{MLE}}(b) \right].$$

- Problem:** The joint estimation of $(\theta, \gamma_1, \dots, \gamma_d)$ is often **computationally expensive** for high-dimensional data.



Two-Stage MLE

The **joint log-likelihood function** for the sample is given by:

$$\ell(\boldsymbol{\eta}) = \sum_{i=1}^n \log c(F_1(X_{i,1}; \gamma_1), \dots, F_d(X_{i,d}; \gamma_d); \boldsymbol{\theta}) + \sum_{i=1}^n \sum_{l=1}^d \log f_l(X_{i,l}; \gamma_l),$$

where $\boldsymbol{\eta} = (\boldsymbol{\theta}, \gamma_1, \dots, \gamma_d)$.

Stage 1 (Marginal Estimation): For each $l = 1, \dots, d$, obtain the marginal maximum likelihood estimate $\tilde{\gamma}_l$ by maximizing the marginal log-likelihood with respect to γ_l :

$$\tilde{\gamma}_l = \arg \max_{\gamma_l} \sum_{i=1}^n \log f_l(X_{i,l}; \gamma_l).$$



Two-Stage MLE

Stage 2 (Copula Estimation): Plug in $\tilde{\gamma} = (\tilde{\gamma}_1, \dots, \tilde{\gamma}_d)$ from Stage 1, to get

$$\ell(\boldsymbol{\theta}, \tilde{\gamma}) = \sum_{i=1}^n \log c(F_1(X_{i,1}; \tilde{\gamma}_1), \dots, F_d(X_{i,d}; \tilde{\gamma}_d); \boldsymbol{\theta}) + \sum_{i=1}^n \sum_{l=1}^d \log f_l(X_{i,l}; \tilde{\gamma}_l).$$

Then, $\boldsymbol{\theta}$ is estimated by maximizing $\ell(\boldsymbol{\theta}, \tilde{\gamma})$ with respect to $\boldsymbol{\theta}$, yielding

$$\tilde{\boldsymbol{\theta}} = \arg \max_{\boldsymbol{\theta}} \ell(\boldsymbol{\theta}, \tilde{\gamma}).$$

The final two-stage maximum likelihood estimate for the entire model is

$$\tilde{\boldsymbol{\eta}} = (\tilde{\boldsymbol{\theta}}, \tilde{\gamma}).$$

Remark

If one wishes to rank the fitted models using these two-stage estimates $\tilde{\boldsymbol{\eta}}$, the standard TIC_{MLE} formula cannot be applied.



Two-Stage MLE based Information Criteria

- In Ko and Hjort (2019), the copula information criterion $\text{CIC}_{2\text{MLE}}$ was derived for two-stage MLE.
- This criterion is an analogue of the TIC_{MLE} in the sense that it **does not assume** that $\mathcal{F}_b$ includes f° :

$$\text{CIC}_{2\text{MLE}}(b) = 2 \left[\ell(\tilde{\eta}_b) - \widetilde{\text{bias}}_{2\text{MLE}}(b) \right].$$

- The authors also showed that if $\mathcal{F}_b$ includes f° , the $\text{CIC}_{2\text{MLE}}$ simply reduces to:

$$\text{AIC}_{2\text{MLE}}(b) = 2 [\ell(\tilde{\eta}_b) - \dim(\eta_b)].$$

- This means that the naive adaptation of AIC_{MLE} to the two-stage MLE is theoretically justified, assuming $f^\circ \in \mathcal{F}_b$.
- To investigate the behavior of the $\text{CIC}_{2\text{MLE}}$ and the $\text{AIC}_{2\text{MLE}}$, we replicate the simulation study presented in Ko and Hjort (2019).



Simulation Objectives

The primary objectives are to:

- Examine the similarity/dissimilarity between CIC_{2MLE} and AIC_{2MLE} ;
- Determine if models with higher CIC_{2MLE} or AIC_{2MLE} values are truly "better" in practice;
- Compare the behavior of the bias-correction term $\widetilde{bias}_{2MLE}$ and the naive penalty $\dim(\boldsymbol{\eta})$.



Data-Generating Process

- We assume the true data-generating density f° is a 4-dimensional pdf ($d = 4$).
- **Dependence:** Governed by a Gumbel copula with parameter $\theta^\circ = 3$.
- **Marginal Distributions:**
 - Margin 1: Weibull with $\gamma_1^\circ = 1.5$ (shape) and $\gamma_2^\circ = 4$ (scale).
 - Margin 2: Weibull with $\gamma_3^\circ = 2$ (shape) and $\gamma_4^\circ = 3$ (scale).
 - Margin 3: Gamma with $\gamma_5^\circ = 2$ (shape) and $\gamma_6^\circ = 1$ (rate).
 - Margin 4: Gamma with $\gamma_7^\circ = 3$ (shape) and $\gamma_8^\circ = 1$ (rate).

We generate $n = 1000$ independent observations per simulation run from this true model.



Candidate Copulas and Margins

To evaluate the criteria, we define a set of $B = 324$ fully parametric models based on all possible combinations of the candidates in Table 1.

Candidates	
Copula	Gumbel, Gaussian, Frank, Survival Clayton
Margin 1	Weibull, Gamma, Log-normal
Margin 2	Weibull, Gamma, Normal
Margin 3	Gamma, Weibull, Log-normal
Margin 4	Gamma, Weibull, Log-normal

Table 1: List of the candidate copulas and margins.



Evaluation Metric: Joint Tail Probability

Motivation: Theory vs. Practice

- $\text{CIC}_{2\text{MLE}}$ minimizes estimated **Kullback-Leibler divergence**.
- While theoretically sounds good, does it really identify models that are useful in practice (e.g., in finance or hydrology)?

Practical Utility Metric: Multivariate Extremes

- We focus on the probability of simultaneous extreme events:

$$\text{TP}^\circ = P\left(X_1 > q_{0.7}^{(1)}, \dots, X_4 > q_{0.7}^{(4)}\right) = P\left(F_1^\circ(X_1) > 0.7, \dots, F_4^\circ(X_4) > 0.7\right), \quad (1)$$

where $q_{0.7}^{(l)} = (F_l^\circ)^{-1}(0.7)$ are the true 0.7-quantiles.

Simulation Setup

- For each of the **324 candidate models**, we estimate this probability.
- **Performance Measure:** Mean Squared Error (MSE) relative to the true probability (1) across 1000 Monte-Carlo replications.



Estimation of the Joint Tail Probability

- For each of the 324 candidate models, we obtain the two-stage MLEs $\left(\tilde{\boldsymbol{\theta}}^{(m)}, \tilde{\gamma}_1^{(m)}, \tilde{\gamma}_2^{(m)}, \tilde{\gamma}_3^{(m)}, \tilde{\gamma}_4^{(m)}\right)$ in each Monte Carlo replication $m = 1, \dots, M$.
- We map the true thresholds $q_{0.7}^{(l)}$ to the unit interval $[0, 1]$ using the estimated marginal cdfs:

$$\tilde{u}_l^{(m)} = F_l \left(q_{0.7}^{(l)}; \tilde{\gamma}_l^{(m)} \right) \quad \text{for } l = 1, \dots, 4.$$

- The joint tail probability (1) is then estimated as:

$$\widetilde{\text{TP}}_m = \int_{\tilde{u}_1^{(m)}}^1 \int_{\tilde{u}_2^{(m)}}^1 \int_{\tilde{u}_3^{(m)}}^1 \int_{\tilde{u}_4^{(m)}}^1 c \left(u_1, u_2, u_3, u_4; \tilde{\boldsymbol{\theta}}^{(m)} \right) du_1 du_2 du_3 du_4.$$

- Finally, we calculate the MSE across $M = 1000$ replications relative to the true joint tail probability TP° :

$$\text{MSE} = \frac{1}{M} \sum_{m=1}^M \left(\widetilde{\text{TP}}_m - \text{TP}^\circ \right)^2.$$



CIC_{2MLE} vs AIC_{2MLE}

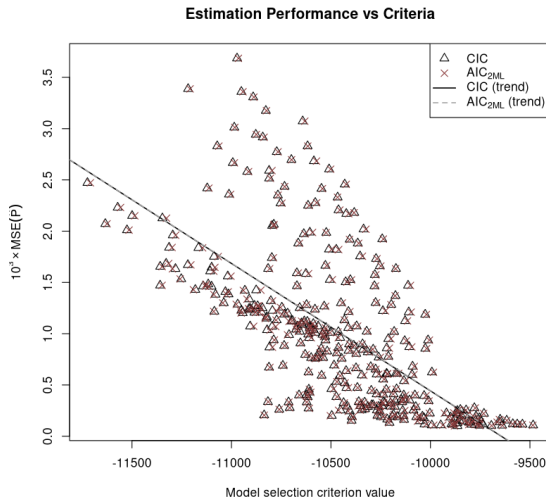


Figure 1: The Mean Squared Error of the estimated joint tail probability (1) versus the average values of the CIC_{2MLE} and the AIC_{2MLE} for all 324 candidate models across 1000 Monte Carlo replications.



Summary: Fully Parametric Approach

Key Findings for 2-Stage MLE:

- CIC_{2MLE} penalizes misspecified models more heavily through its complex bias-correction term. However, this conceptual superiority provides little practical advantage for large sample sizes (e.g., $n = 1000$).
- For smaller sample sizes, the empirically estimated $\widetilde{bias}_{2MLE}$ could potentially dominate the likelihood term, leading to increased variability in the CIC_{2MLE} performance.

Drawbacks of the fully parametric approach:

- The estimates of the dependence parameters θ are highly sensitive to the choice of the marginal models.
- Misspecified marginals can severely distort the estimates of the copula parameters θ .

How do we select copula models when the true nature of the marginals is completely unknown?

→ **Part II: The Semiparametric Approach**



Semiparametric Approach

- Assume that we are only interested in the reliable estimation of the dependence parameters θ , and we don't want to make any parametric assumptions about marginals.
- Instead of fitting models to the **independent observations** $\mathcal{X}_n \subset \mathbb{R}^d$, we fit only copula models to the **dependent pseudo-observations** ${}^p\mathcal{X}_n \subset [0, 1]^d$.
- The transformation $\mathcal{X}_n \mapsto {}^p\mathcal{X}_n$ is defined by the function

$$\mathbf{F}_{n,\perp}(x_1, \dots, x_d) = (F_{n,1}(x_1), \dots, F_{n,d}(x_d)),$$

where $F_{n,l}$ is the $\frac{n}{n+1}$ -rescaled empirical cdf of the l -th marginal, for $l = 1, \dots, d$.

- The corresponding pseudo-observations ${}^p\mathcal{X}_n = \{{}^p\mathbf{X}_i\}_{i=1}^n$ are then given by ${}^p\mathbf{X}_i = \mathbf{F}_{n,\perp}(\mathbf{X}_i)$, for all $i = 1, \dots, n$.



Maximum Pseudo-Likelihood Estimator (MPLE)

- The **pseudo-log-likelihood** is then defined as

$${}^P\ell(\boldsymbol{\theta}) = \sum_{i=1}^n \log[c({}^P\mathbf{X}_i; \boldsymbol{\theta})].$$

- The **maximum pseudo-likelihood estimator (MPLE)** is given by

$${}^P\hat{\boldsymbol{\theta}} = \operatorname{argmax}_{\boldsymbol{\theta} \in \Theta} {}^P\ell(\boldsymbol{\theta}).$$

- Genest et al. (1995) showed that this maximizer exhibits the same asymptotic behavior as the classical MLE. Specifically, it is consistent:

$${}^P\hat{\boldsymbol{\theta}} \xrightarrow[n \rightarrow \infty]{P} \operatorname{arg} \min_{\boldsymbol{\theta} \in \Theta} \int_{[0,1]^d} \log \left[\frac{c^\circ(\mathbf{u})}{c(\mathbf{u}; \boldsymbol{\theta})} \right] dC^\circ(\mathbf{u}).$$



Naive Adaptation of AIC to the Semiparametric Case

- Consider $B \in \mathbb{N}$ candidate parametric copula families:

$$\mathcal{C}_b = \{c(\cdot; \boldsymbol{\theta}_b) : \boldsymbol{\theta}_b \in \Theta_b\} \quad \text{for } b = 1, \dots, B.$$

- Using MPLE, we fit the candidate copula families to our data to identify the most suitable model.
- A **naive adaptation** of AIC to the semiparametric case yields

$${}^p\text{AIC} = 2 \cdot \left[{}^p\ell \left({}^p\hat{\boldsymbol{\theta}} \right) - \dim(\boldsymbol{\theta}) \right].$$

- When using the ${}^p\text{AIC}$, it was initially hoped that because the MPLE heuristically resembled the MLE, there would be a continuous relationship between the ${}^p\text{AIC}$ and the classical AIC_{MLE} .



Problems with p AIC

- p AIC ignores the noise introduced when transforming data into pseudo-observations.
- Grønneberg and Hjort (2014) adapted classical AIC_{MLE} arguments to the semiparametric setting, deriving an information criterion valid only under **highly restrictive conditions**.
- However, these **conditions are not satisfied** by popular copula models.
- Furthermore, they proved a much stronger result: a generally applicable **AIC-like formula does not even exist** for many popular copula models under MPLE.
- **Conclusion:** In the copula context, **when viewing model selection from a loss-function perspective**, standard penalty-based criteria are simply not suitable for pseudo-likelihood estimation.
- Nevertheless, p AIC remains widely used due to its computational simplicity.



Equivalence Between TIC_{MLE} and xv_1

The Classical Parametric Setting (MLE)

- It is well known that TIC_{MLE} is asymptotically equivalent to leave-one-out cross-validation xv_1 :

$$\text{TIC}_{\text{MLE}} = 2n \cdot xv_1 + o_P(1), \quad (2)$$

where $xv_1 = \frac{1}{n} \sum_{i=1}^n \log \left[f \left(\mathbf{X}_i; \hat{\boldsymbol{\eta}}_{(-i)} \right) \right]$, and $\hat{\boldsymbol{\eta}}_{(-i)}$ is the MLE based on the sample $\mathcal{X}_n$ without the i -th observation $\mathbf{X}_i$.

The Semiparametric Setting MPLE

- However, in Section 4 of Grønneberg and Hjort (2014), it was shown that in the case of the MPLE, an equivalence analogous to (2) does not hold, which provides a further contrast between MPLE-based and MLE-based estimation.



Leave-One-Out Copula Information Criterion

Motivated by a **prediction perspective**, the leave-one-out cross-validation information criterion is defined as:

$$p_{\text{XV}_1} = \frac{1}{n} \sum_{i=1}^n \log \left[c \left(\mathbf{F}_{n,\perp,(-i)}(\mathbf{X}_i); {}^p\hat{\boldsymbol{\theta}}_{(-i)} \right) \right], \text{ where}$$

- $\mathbf{F}_{n,\perp,(-i)}(x_1, \dots, x_d) = (F_{(-i),1}(x_1), \dots, F_{(-i),d}(x_d))$, where $F_{(-i),l}$ is the $\frac{n-1}{n}$ -rescaled empirical cdf of the l -th marginal, computed from the sample $\mathcal{X}_n$ excluding $\mathbf{X}_i$, for $l = 1, \dots, d$,
- ${}^p\hat{\boldsymbol{\theta}}_{(-i)} = \operatorname{argmax}_{\boldsymbol{\theta} \in \Theta} \sum_{j \neq i} \log [c(\mathbf{F}_{n,\perp,(-i)}(\mathbf{X}_j); \boldsymbol{\theta})]$.



Bias-Corrected Approximation of p_{xv_1}

However, since computing p_{xv_1} is **computationally expensive**, the authors recommend using its first-order asymptotic approximation, xv_{CIC} , defined as:

$$xv_{CIC} = 2 \cdot \left[p\ell \left(\widehat{p\theta} \right) - \widehat{bias}_{CIC} \right],$$

where $\widehat{bias}_{CIC}$ is a bias-correction term (see Section 4 of Grønneberg and Hjort (2014) for details).

Paradox: Although p_{xv_1} and xv_{CIC} are both **theoretically valid** information criteria, simulation studies (see Jordanger and Tjøstheim (2014)) demonstrate that **neither can outperform** $pAIC$ — a criterion that lacks theoretical justification.



Alternative Cross-Validation Approaches

- In the context of **linear model selection**, Shao (1993) showed that the optimal selection procedure is leave- n_v -out cross-validation, where the **the validation set size** n_v is of the same order as the full sample size n , that is, $n_v/n \rightarrow 1$ as $n \rightarrow \infty$.
- During the summer of 2025, I attempted to “naively” adapt Shao’s approach to the copula setting. The proposed method was still **unable to beat** the well-known ^pAIC .
- As an obvious **cross-validation-based** alternative, one can try using K -fold cross-validation with the standard values $K = 5, 10$.
- Are there **other possible choices** for K , and what is the **motivation** behind them?



Choosing K_n : A Data-Driven Approach

Basic properties of leave-one-out cross-validation P_{XV_1} :

- 1 $n_v = 1$,
 - 2 $\frac{n_v}{n} \xrightarrow{n \rightarrow \infty} 0$,
 - 3 $K_n = \frac{n}{n_v} = \frac{n}{1} = n$ [P_{XV_1} is a special case of K_n -fold cross-validation].
- Since P_{XV_1} is a **theoretically justified** information criterion and is a **special case** of $P_{\text{XV}_{K_n}}$, one might wonder how $P_{\text{XV}_{K_n}}$ would perform for values of K_n such that the property (2) still holds.
 - In other words, $P_{\text{XV}_{K_n}}$ aims to imitate P_{XV_1} in the sense that the validation set remains small, while avoiding the overly "conservative" choice $n_v = 1$.
 - **The question of interest** is whether $P_{\text{XV}_{K_n}}$ can outperform P_{AIC} .



K_n -Fold Cross-Validation

We randomly partition the training data $\mathcal{X}_n$ into K_n disjoint subsets of equal size $n_v = \frac{n}{K_n}$. Let $\mathcal{I}_k$ denote set of indices belonging to the k -th fold for $k = 1, \dots, K_n$.

$$p_{\text{XV}_{K_n}} = \frac{1}{n} \sum_{k=1}^{K_n} \sum_{i \in \mathcal{I}_k} \log \left[c \left(\mathbf{F}_{n, \perp, (-\mathcal{I}_k)}(\mathbf{X}_i); \hat{\theta}_{(-\mathcal{I}_k)} \right) \right], \text{ where}$$

- $\hat{\theta}_{(-\mathcal{I}_k)} = \operatorname{argmax}_{\theta \in \Theta} \sum_{j \notin \mathcal{I}_k} \log \left[c \left(\mathbf{F}_{n, \perp, (-\mathcal{I}_k)}(\mathbf{X}_j); \theta \right) \right],$
- $\mathbf{F}_{n, \perp, (-\mathcal{I}_k)}(x_1, \dots, x_d) = (F_{(-\mathcal{I}_k), 1}(x_1), \dots, F_{(-\mathcal{I}_k), d}(x_d)),$ where $F_{(-\mathcal{I}_k), l}$ is the $\frac{n-n_v}{n}$ -rescaled empirical cdf of the l -th marginal, computed from the sample $\mathcal{X}_n$ excluding $\{\mathbf{X}_j : j \in \mathcal{I}_k\}$, for $l = 1, \dots, d.$

In our simulation study, we used both fixed and data-driven choices for the number of folds:

$$K_n \in \left\{ 2, 5, 10, \left\lfloor \frac{\sqrt{n}}{\log(n)} \right\rfloor, \lfloor \sqrt{n} \rfloor, \left\lfloor \frac{n}{\log(n)} \right\rfloor \right\}.$$



Summary of MPLE-based Information Criteria

In the semiparametric setting of copula model selection, we consider the following four information criteria:

- 1 Naive Akaike Information Criterion: ^pAIC (**not valid, easy to compute**)
- 2 Leave-One-Out Cross-Validation: $^p\text{xv}_1$ (**valid, computationally expensive**)
- 3 Approximate Leave-One-Out Criterion: xv_{CIC} (**valid, moderately expensive to compute**)
- 4 K_n -fold Cross-Validation: $^p\text{xv}_{K_n}$ (**?, computationally expensive**)

In the study Jordanger and Tjøstheim (2014), the authors compared xv_{CIC} and ^pAIC .



Setup of the Simulation Study

The simulation study is based on the following settings:

- We constrain our analysis to bivariate distributions ($d = 2$).
- We simulate true data-generating models from five widely used one-parameter copula families:

Clayton, Gumbel, Joe, Frank, Gaussian.

- Each copula was parametrized using different values of Kendall's tau

$$\tau \in \{0.05, 0.1, 0.15, 0.2, 0.25, 0.5, 0.75\}.$$

- Considered sample sizes $n \in \{100, 200, 400\}$.
- For each combination of the true model, τ and n , we conducted 1000 Monte Carlo replications.

During each MC replication, all five candidate copula families are fitted to the generated data using the MPLE procedure.



Evaluation Metric: Hit Rates

Objective: Measure how often a criterion successfully identifies the true data-generating copula c° .

Step 1: Model Selection

- In each replication m , the criterion assigns a numerical score $S_m(\text{crit}, c)$ to the fitted candidate family c .
- The selected model is the one that maximizes this score:

$$c_{m,\text{crit}}^{\text{selected}} = \arg \max_{c \in \mathcal{C}_{\text{cand}}} S_m(\text{crit}, c).$$

Step 2: Empirical Hit Rate (HR)

- We define an indicator $I_m(\text{crit}) = \mathbb{I}(c_{m,\text{crit}}^{\text{selected}} = c^\circ)$.
- The Hit Rate is the sample mean across $M = 1000$ replications:

$$\text{HR}(\text{crit}) = \frac{1}{M} \sum_{m=1}^M I_m(\text{crit}).$$



Evaluation Metric: Hit Rates

Step 3: Asymptotic Confidence Intervals

- Treating the indicators as Bernoulli trials, we use the CLT to construct 95% CIs:

$$\text{HR}(\text{crit}) \pm z_{0.975} \times \frac{s_{\text{crit}}}{\sqrt{M}},$$

where s_{crit} denotes the sample standard deviation of the indicator variables $\{I_m(\text{crit})\}_{m=1}^M$.

- *Purpose:* This allows us to rigorously determine if performance differences between criteria are statistically significant.



Results and Discussion: Identification of the True Copula

IC	Clayton	Gumbel	Joe	Frank	Gaussian
P_{AIC}	55.50 $\pm$ 3.080	13.20 $\pm$ 2.100	51.30 $\pm$ 3.100	32.80 $\pm$ 2.910	18.00 $\pm$ 2.380
P_{XV_1}	55.50 $\pm$ 3.080	13.20 $\pm$ 2.100	51.30 $\pm$ 3.100	32.80 $\pm$ 2.910	18.00 $\pm$ 2.380
xv_{CIC}	55.70 $\pm$ 3.080	10.60 $\pm$ 1.910	53.60 $\pm$ 3.090	29.20 $\pm$ 2.820	18.40 $\pm$ 2.400
$K = 2$	40.00 $\pm$ 3.040	13.80 $\pm$ 2.140	43.90 $\pm$ 3.080	23.80 $\pm$ 2.640	16.20 $\pm$ 2.280
$K = 5$	38.50 $\pm$ 3.020	18.00 $\pm$ 2.380	48.20 $\pm$ 3.100	22.30 $\pm$ 2.580	13.80 $\pm$ 2.140
$K = 10$	38.80 $\pm$ 3.020	21.50 $\pm$ 2.550	50.60 $\pm$ 3.100	25.90 $\pm$ 2.720	13.90 $\pm$ 2.150
$K_n = \left\lfloor \frac{n}{\log(n)} \right\rfloor$	36.70 $\pm$ 2.990	24.40 $\pm$ 2.660	52.10 $\pm$ 3.100	24.60 $\pm$ 2.670	15.40 $\pm$ 2.240
$K_n = \sqrt{n}$	38.60 $\pm$ 3.020	22.40 $\pm$ 2.590	50.90 $\pm$ 3.100	24.50 $\pm$ 2.670	15.40 $\pm$ 2.240
$K_n = \left\lfloor \frac{\sqrt{n}}{\log(n)} \right\rfloor$	39.90 $\pm$ 3.040	15.30 $\pm$ 2.230	43.10 $\pm$ 3.070	23.70 $\pm$ 2.640	14.30 $\pm$ 2.170

Table 2: Hit rates ($n = 200$, $\tau = 0.05$), with 95 % confidence intervals (all values multiplied by 100).



Results and Discussion: Identification of the True Copula

IC	Clayton	Gumbel	Joe	Frank	Gaussian
P_{AIC}	97.90 $\pm$ 0.890	68.70 $\pm$ 2.880	86.40 $\pm$ 2.130	75.50 $\pm$ 2.670	67.80 $\pm$ 2.900
P_{XV_1}	97.90 $\pm$ 0.890	68.70 $\pm$ 2.880	86.40 $\pm$ 2.130	75.50 $\pm$ 2.670	67.80 $\pm$ 2.900
xv_{CIC}	96.10 $\pm$ 1.200	65.90 $\pm$ 2.940	88.60 $\pm$ 1.970	77.40 $\pm$ 2.590	67.00 $\pm$ 2.920
$K = 2$	96.00 $\pm$ 1.220	64.70 $\pm$ 2.960	83.20 $\pm$ 2.320	69.50 $\pm$ 2.860	61.90 $\pm$ 3.010
$K = 5$	97.70 $\pm$ 0.930	68.20 $\pm$ 2.890	85.50 $\pm$ 2.180	75.80 $\pm$ 2.660	67.50 $\pm$ 2.900
$K = 10$	97.60 $\pm$ 0.950	69.40 $\pm$ 2.860	84.80 $\pm$ 2.230	75.70 $\pm$ 2.660	66.90 $\pm$ 2.920
$K_n = \left\lfloor \frac{n}{\log(n)} \right\rfloor$	97.70 $\pm$ 0.930	69.20 $\pm$ 2.860	85.70 $\pm$ 2.170	77.00 $\pm$ 2.610	66.90 $\pm$ 2.920
$K_n = \sqrt{n}$	98.00 $\pm$ 0.870	68.70 $\pm$ 2.880	86.20 $\pm$ 2.140	77.40 $\pm$ 2.590	67.50 $\pm$ 2.900
$K_n = \left\lfloor \frac{\sqrt{n}}{\log(n)} \right\rfloor$	97.20 $\pm$ 1.020	67.90 $\pm$ 2.900	85.20 $\pm$ 2.200	75.60 $\pm$ 2.660	64.90 $\pm$ 2.960

Table 3: Hit rates ($n = 400$, $\tau = 0.25$), with 95 % confidence intervals (all values multiplied by 100).



Coincidence Percentages

Objective: Quantify how often two distinct criteria (crit_1 and crit_2) agree, regardless of whether their choice is the true copula model.

Step 1: The Coincidence Indicator

- In replication m , we check if both criteria choose the exact same candidate copula family:

$$J_m(\text{crit}_1, \text{crit}_2) = \mathbb{I} \left(c_{m, \text{crit}_1}^{\text{selected}} = c_{m, \text{crit}_2}^{\text{selected}} \right).$$

Step 2: Empirical Coincidence Percentage (CP)

- The CP is the sample mean of these agreements across all M replications:

$$\text{CP}(\text{crit}_1, \text{crit}_2) = \frac{1}{M} \sum_{m=1}^M J_m(\text{crit}_1, \text{crit}_2).$$



Results and Discussion: Criteria Agreement

	$\tau = 0.05$	$\tau = 0.1$	$\tau = 0.15$	$\tau = 0.2$	$\tau = 0.25$
100	99.74	99.92	99.90	99.96	99.94
200	99.92	99.98	99.98	99.98	99.98
400	100.00	99.98	100.00	100.00	100.00

Table 4: Coincidence of PAIC and $^P_{xv_1}$ (all values multiplied by 100).

	$\tau = 0.05$	$\tau = 0.1$	$\tau = 0.15$	$\tau = 0.2$	$\tau = 0.25$
100	79.26	86.46	89.10	89.34	90.34
200	86.88	91.38	92.90	94.10	94.72
400	92.52	95.46	96.38	96.60	97.26

Table 5: Coincidence of PAIC and xv_{CIC} (all values multiplied by 100).



Summary: Semiparametric Model Selection

- **The Paradox of $P\text{AIC}$:** Despite lacking theoretical justification, it performs almost indistinguishably from the rigorously leave-one-out cross-validation P_{xv_1} .
- **Performance of the xv_{CIC} :**
 - Serves as an excellent asymptotic approximation of P_{xv_1} in large samples.
 - Deviates in small samples because its complex bias-correction terms introduce finite-sample variation.
- **Evaluation of K_n -Fold Cross-Validation $P_{\text{xv}_{K_n}}$:**
 - **Goal:** Employ data-driven folds to mimic the asymptotic properties of P_{xv_1} while reducing its computational cost.
 - **Conclusion:** Whether using fixed or dynamic choices for K_n , $P_{\text{xv}_{K_n}}$ does **not** yield a statistically superior criterion compared to existing methods.



Conclusion: Guidelines for Copula Model Selection

Step 1: The Marginal Assumption

- *Margins known (e.g., via domain experts)?* → Fully Parametric Strategy (Two-Stage MLE).
- *Margins unknown?* → Semiparametric Strategy (MPLE).

Step 2: Fully Parametric Criteria (AIC_{2MLE} vs. CIC_{2MLE})

- **Large samples:** AIC_{2MLE} is safe to use even when the true copula is misspecified. It is theoretically justified and performs identically to CIC_{2MLE} .
- **Small samples:** One should still be careful with CIC_{2MLE} . Its asymptotic bias-correction terms can become unstable.

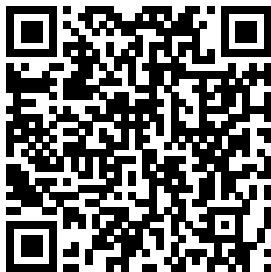
Step 3: Semiparametric Criteria (PAIC vs. xv_{CIC})

- In contrast to the parametric case, PAIC completely lacks theoretical justification.
- Despite this, PAIC empirically matches the performance of exact cross-validation Pxv_1 .



Thank you for your attention!

FULL SIMULATION STUDY



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